

Introduction & Motivation

Large language models can answer complex questions, but their reasoning may contain unsupported or hallucinated facts. Knowledge graphs provide structured factual evidence, and Graph-Constrained Reasoning uses KG-Tries to force generated reasoning paths to be valid in the graph. However, complex KGQA requires more than valid paths. Many questions involve multiple constraints, temporal filters, numeric conditions, or superlative reasoning. At the same time, token-level constrained decoding is expensive and its beam scores do not always reflect answer correctness. This project studies two complementary ways to improve KG-grounded LLM reasoning: **DVI** improves the reasoning state by explicitly decomposing and intersecting constraints. **GraphLite** improves the verifier by replacing token-level path generation with faster entity-level relation and candidate scoring.

Problem Statement

GCR grounds LLM reasoning by ensuring that generated paths exist in the knowledge graph. But two limitations remain:

1. Constraint satisfaction gap

A KG-valid path does not guarantee that the final answer satisfies all constraints in a complex question.

2. Verifier efficiency gap

KG-LLM constrained decoding is slow, and token-level generation scores are not direct correctness scores for candidate answers.

Therefore, this project asks:

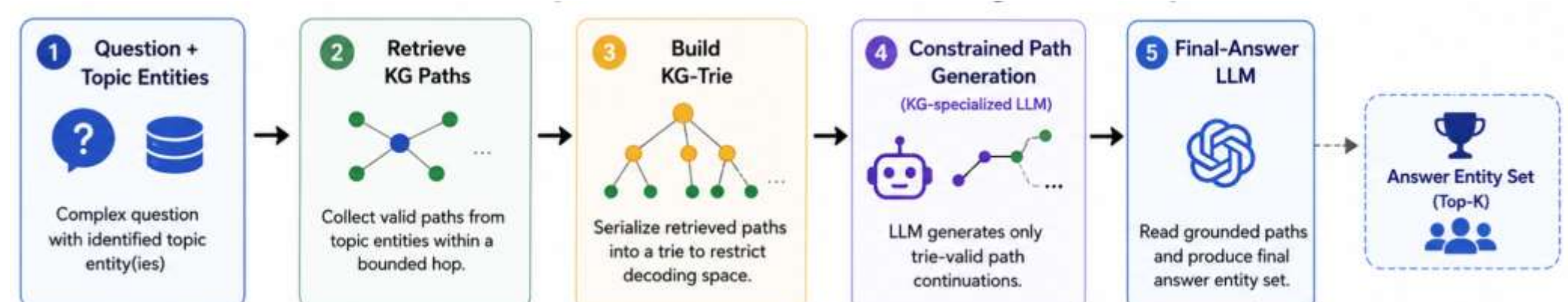
How can KGQA move from path-valid reasoning toward constraint-satisfying and efficient entity-level verification?

Baseline: Graph-Constrained Reasoning (GCR) [1]

GCR grounds LLM reasoning by constraining path generation with a KG-Trie.

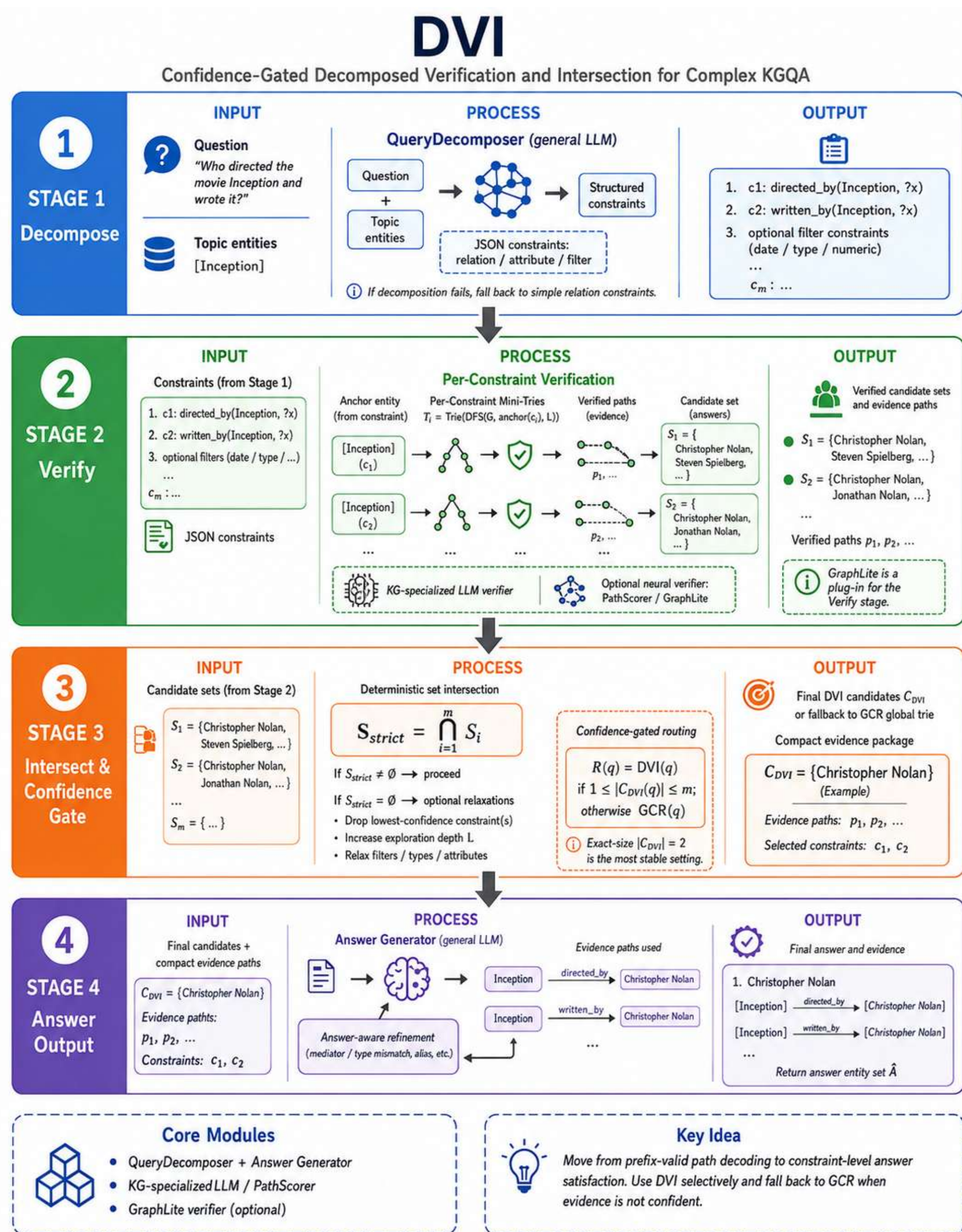
Given topic entities, GCR retrieves valid KG paths, serializes them into a trie, and forces the KG-specialized LLM to generate only trie-valid path continuations. A general LLM then reads the grounded paths and produces final answers.

Limitation: GCR ensures path validity, but it does not explicitly perform candidate intersection, temporal/numeric filtering, or superlative comparison. It also relies on token-level constrained decoding, which can be slow.



Methodology

DVI addresses the reasoning-quality problem: it decomposes complex questions into constraints, verifies each constraint with KG evidence, intersects candidate sets, and uses a confidence gate to fall back to GCR when DVI evidence is unreliable.



Result

➤ Result 1: Confidence-Gated DVI slightly improves GCR when used selectively

Method	KG model	N	Acc./Rec.	Hit	Hit@1	ΔHit@1	F1	ΔF1	Prec.	Time(s)
GCR baseline	Llama-3.1-8B	3531	59.01	64.34	55.88	—	54.55	—	54.87	—
Raw DVI clean top-1	Llama-3.1-8B	3520	57.71	64.32	54.18	-1.70	52.57	-1.98	53.42	12.72
Gated DVI, C ≤ 1	Llama-3.1-8B	3531	59.03	64.43	54.80	-1.08	54.33	-0.22	54.55	8.54
Gated DVI, C ≤ 2	Llama-3.1-8B	3531	59.26	64.68	56.22	+0.34	54.63	+0.08	54.90	6.58
Gated DVI, C ≤ 2	Llama-3.1-8B	3531	59.28	64.77	55.14	-0.74	54.41	-0.14	54.59	10.29
Gated DVI, C ≤ 3	Llama-3.1-8B	3531	58.92	64.40	54.63	-1.25	53.87	-0.68	54.05	11.27
GCR baseline	Llama-2-7B	3531	60.51	66.07	57.12	—	54.50	—	54.29	6.27
Raw DVI clean top-1	Llama-2-7B	3520	57.76	64.35	53.86	-3.26	52.31	-2.19	53.02	7.92
Gated DVI, C ≤ 1	Llama-2-7B	3531	60.40	66.02	55.88	-1.24	54.38	-0.12	54.15	6.89
Gated DVI, C ≤ 2	Llama-2-7B	3531	60.56	66.27	56.92	-0.20	54.51	+0.01	54.42	6.53
Gated DVI, C ≤ 2	Llama-2-7B	3531	60.45	66.21	55.68	-1.44	54.40	-0.10	54.28	7.16
Gated DVI, C ≤ 3	Llama-2-7B	3531	59.71	65.59	54.91	-2.21	53.84	-0.66	53.88	7.34

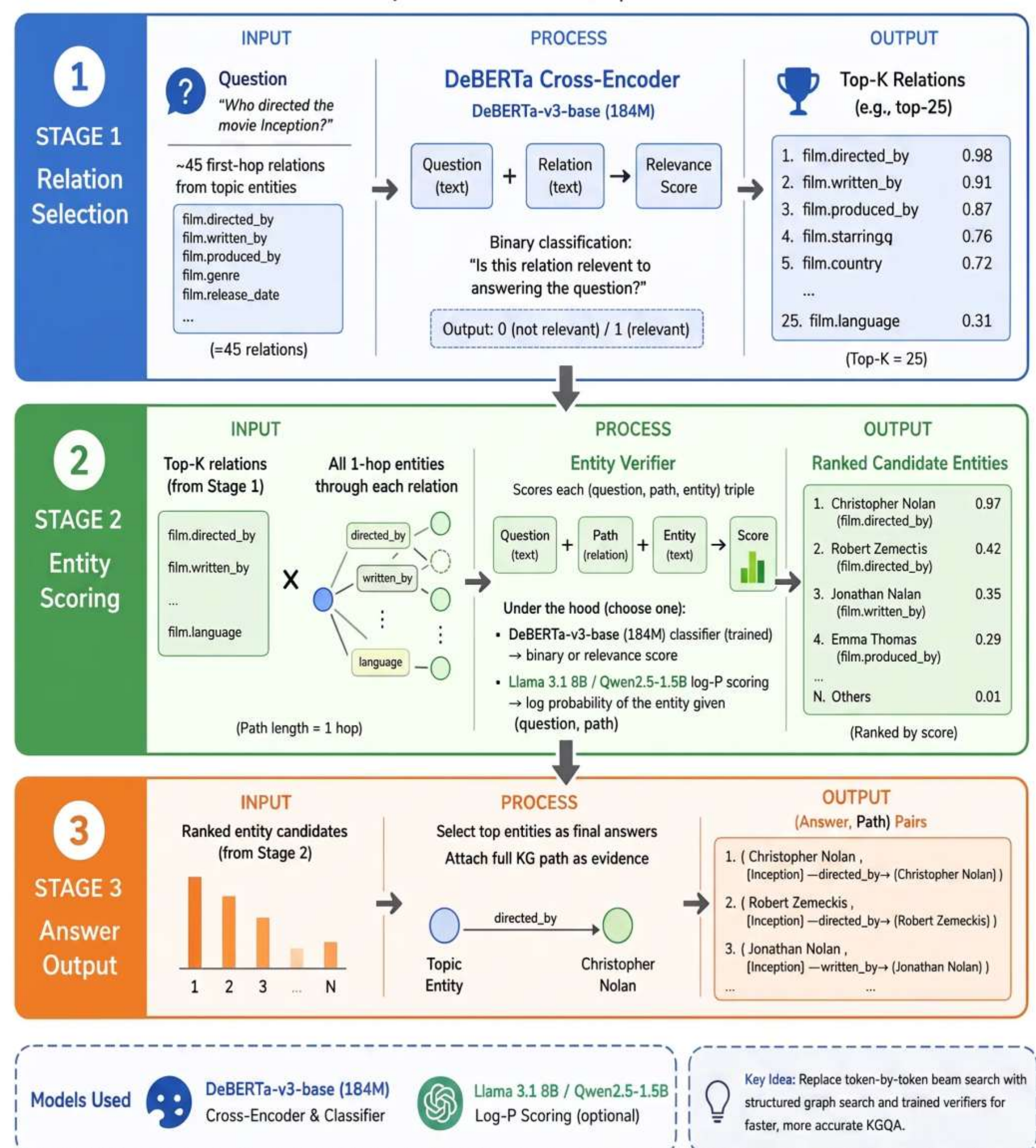
➤ Result 2: GraphLite is faster, but still below KG-LLM verification in QA quality

Method	Verifier	N	Hit	ΔHit	Hit@1	ΔHit@1	F1	ΔF1	Time(s)
GCR baseline	Llama-3.1 KG-LLM	3531	64.34	—	55.88	—	54.55	—	—
Raw DVI clean top-1	KG-LLM verification	3520	64.32	-0.02	54.18	-1.70	52.57	-1.98	12.72
PathScorer DVI	Bi-encoder + cross-encoder	3520	56.19	-8.15	45.09	-10.79	42.14	-12.41	3.09
GraphLite entity-level verifier	Relation/entity scorer	3531	59.25	-5.09	—	—	44.90	-9.65	~1.50

GraphLite addresses the verifier-efficiency problem: it replaces token-level constrained decoding with structured relation selection and entity-level scoring for faster KGQA verification.

GraphLite

KGQA with Structured Graph Search



➤ Result 3: Oracle routing reveals large untapped headroom

Although hand-designed gating gives small gains, oracle routing raises F1 from about **54.6** to over **61**, showing that DVI and GCR make complementary errors.

KG model	Source	N	Acc./Rec.	Hit@1	F1
Llama-3.1	GCR baseline	3520	59.13	56.05	54.63
Llama-3.1	Raw DVI	3520	57.71	54.18	52.57
Llama-3.1	Oracle-by-F1	3520	65.79	62.36	61.51
Llama-2	GCR baseline	3520	60.63	57.27	54.57
Llama-2	Raw DVI	3520	57.76	53.86	52.31
Llama-2	Oracle-by-F1	3520	66.49	63.35	61.37

Conclusion

This project improves KG-grounded LLM reasoning along two complementary directions: DVI moves reasoning from valid paths to explicit constraint satisfaction, while GraphLite moves verification from slow token-level decoding to faster entity-level scoring.

References

[1] Luo et al. *Graph-constrained Reasoning: Faithful Reasoning on Knowledge Graphs with Large Language Models*. ICML 2025.

Key Findings

- Two bottlenecks:** complex KGQA needs better reasoning and faster verification. **DVI** targets reasoning quality; **GraphLite** targets efficiency.
- KG-valid ≠ answer-correct.** Valid paths still miss intersections, filters, answer types, and superlatives.
- DVI works best selectively.** Raw DVI is brittle, but confidence gating improves GCR when evidence is compact.
- GraphLite is faster.** Entity-level scoring reduces verification time, but quality still trails KG-LLM verification.
- Future: GraphLite-DVI.** Combine GCR grounding, DVI constraint control, GraphLite verification, and learned routing.